Spherical aberrations — Solution

X22 (2022) — English translation

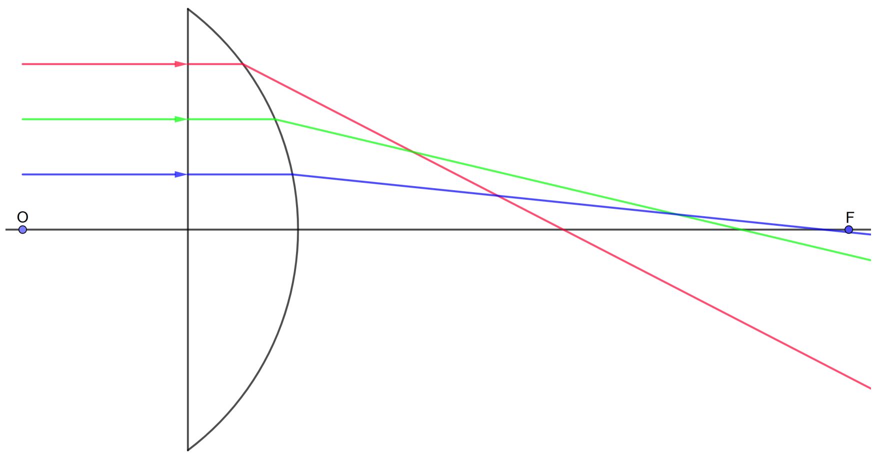
A1 **0.45 pts**

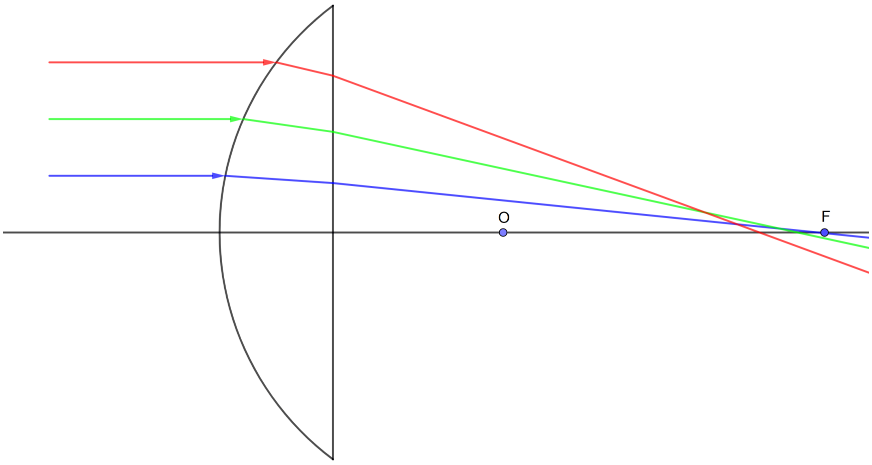
Construct the exact path of rays in a plano-convex and plano-concave lens. The radii of curvature of surfaces is $R = 5\ cm$, the refractive index of glass is $n = 1.5$. Indicate the point F' of convergence (divergence) of the parallel paraxial beam.

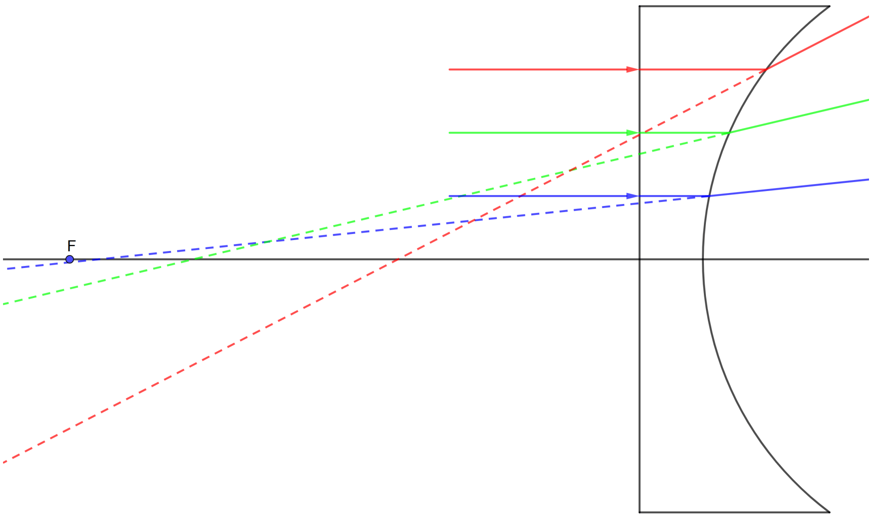
Note that the radius of the lens is 4 cm, the thickness of the lens is 2 cm, and the incident rays are drawn every centimeter from the optical axis of the lens. For a plano-convex (first) and plano-concave (third) lens, refraction occurs on only one surface. The focus position for the first and third lenses is noted using the thin lens formula:

$$F = \frac{R}{n-1} = 10\ cm.$$

Это distance отсчитывается от изогнутой поверхности. For выпукло-плоской (вторая) линзы refraction происходит ссу на двух поверхностях. Чтобы найти положение focusa, учтём, что такая lens является толстой. После первого преломления rayand внутри стекла будут собираться на расстоянии $\frac{n}{n-1}R = 15\ cm$ от изогнутой поверхности, that is на расстоянии 13 cm от плоской поверхности. For/At преломлении на плоской поверхности image окажется в n times ближе, that is на расстоянии 8.67 cm. In total, we get focus на расстоянии 10.67 cm от изогнутой поверхности. Чтобы построить refraction rayей, we use транспортир, линейку and закон Снелиуса: $n_1 \sin\alpha = n_2 \sin\beta$, where α – angle падения, β – angle преломления, n_1 and n_2 are the refractive indices of the media on opposite sides of the boundary.







A2 0.30 pts

Measure the magnitude of the maximum spherical aberration $\delta s'$ for all constructions. Record your results (in mm) on your answer sheet.

Based on the results of the constructions, we measure the maximum value of aberration. For convex-flat, the effect of aberrations is less. Below are the results, accurate to tenths of mm, obtained by calculations.

Answer: \textbfAnswer: Plano-convex $\delta s'_{\max} = -51.8$ mm Convex-flat $\delta s'_{\max} = -11.5$ mm Plano-concave $\delta s'_{\max} = -51.8$ mm

B1 0.25 pts

What is the focal length f' for a thin plano-convex lens?

For a thin lens

$$\frac{1}{f'} = (n-1)\left(\frac{1}{R_1} + \frac{1}{R_2}\right),$$

в нашем crayae $R_1 = \infty$, $R_2 = R$. From here we get the answer.

Answer: \textbfAnswer: $f' = \dfrac{R}{n-1}$

B2 — 1.50 pts

For a given lens, find the dependence of the longitudinal spherical aberration $\delta s'$ on the half-width of the beam h. Express your answer using R, n and h. Write an approximate expression for $\delta s'(h)$ by expanding the answer into a Taylor series of powers h to h^2. Construct a schematic graph of the resulting relationship.

According to geometry (Fig. B2) and Snell's law:

$$n \sin\alpha = \sin(\alpha + \beta).$$

Let us express углы via/through отрезки and refractive index:

$$\alpha = \arcsin\frac{h}{R}, \quad \alpha + \beta = \arcsin\frac{nh}{R}.$$

Distance l вдоль оптической оси от точки преломления до точки пересечения с оптической осью:

$$l = h\,\mathrm{ctg}\beta = h\,\mathrm{ctg}\left(\arcsin\left(\frac{nh}{R}\right) - \arcsin\left(\frac{h}{R}\right)\right).$$

Refraction происходит at the point, смещенной на

$$\delta x = R - \sqrt{R^2 - h^2}.$$

Let us express $\delta s'$:

$$\delta s' = l - \delta x - f'.$$

Substituting l and δx найденные ранее, substituting f' from B1, we get the answer.

Answer: \textbfAnswer: Reply for $\delta s'$

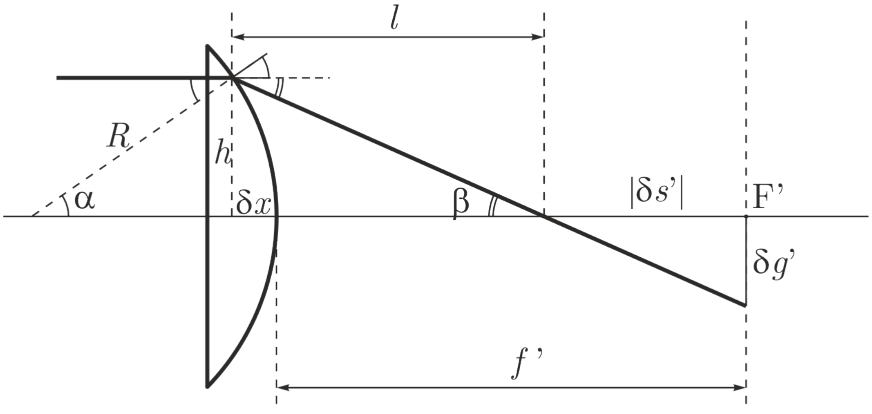
$$\delta s' = \frac{\sqrt{R^2 - n^2h^2}\sqrt{R^2 - h^2} + nh^2}{n\sqrt{R^2 - h^2} - \sqrt{R^2 - n^2h^2}} - \frac{nR}{n-1} + \sqrt{R^2 - h^2}$$

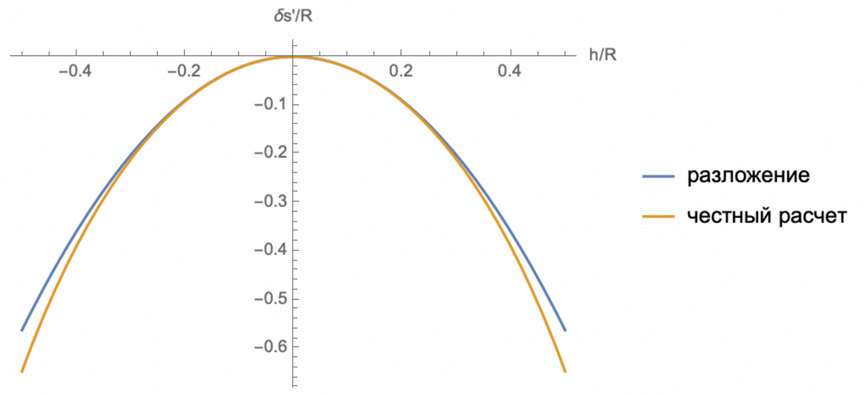
or

$$\delta s' = \frac{n\sqrt{R^2 - h^2n^2} + \sqrt{R^2 - h^2}}{n^2 - 1} - \frac{nR}{n-1} + \sqrt{R^2 - h^2}$$

or

$$h\mathrm{ctg}\left(\arcsin\left(\frac{nh}{R}\right) - \arcsin\left(\frac{h}{R}\right)\right) - \frac{nR}{n-1} + \sqrt{R^2 - h^2}.$$





B3 0.50 pts

Find the value of transverse spherical aberration $\delta g'$. Express your answer in terms of $\delta s'$, f' and h.

From the similarity of triangles (Fig. B2) we obtain

$$\frac{h}{l} = \frac{\delta g'}{|\delta s'|}.$$

Считая аберрации малыми ($l = f'$), we express the answer.

Answer: \textbfAnswer: From similarity of triangles:

$$\delta g' = |\delta s'| \frac{h}{f'}$$

B4 1.50 pts

Let us divide the scattering circle into rings of radii $0.2\delta g', 0.4\delta g', 0.6\delta g', 0.8\delta g'$. From what fraction (in \%) of the transverse area of the incident beam are the rays collected into the central circle (radius $0.2\delta g'$)? From what fraction (in \%) of the transverse area of the incident beam are the rays collected into subsequent rings (between $0.2\delta g'$ and $0.4\delta g'$ and so on, respectively)? What fraction of the area of the scattering circle is occupied by each ring? Tabulate all calculations on your answer sheet. Construct a schematic graph of light intensity versus radial coordinate in the scattering circle.

We count the area fractions of each ring:

$$0 \div 0.2\delta g': \quad 0.2^2 - 0^2 = 0.04 = 4\%;$$

$$0.2\delta g' \div 0.4\delta g': \quad 0.4^2 - 0.2^2 = 0.12 = 12\%;$$

$$0.4\delta g' \div 0.6\delta g': \quad 0.6^2 - 0.4^2 = 0.20 = 20\%;$$

$$0.6\delta g' \div 0.8\delta g': \quad 0.8^2 - 0.6^2 = 0.28 = 28\%;$$

$$0.8\delta g' \div 1.0\delta g': \quad 1.0^2 - 0.8^2 = 0.36 = 36\%;$$

From parta B2 we get $|\delta s'| \sim h^2$. Taking into account part B3 we get, что $\delta g' \sim h^3$. Считаем доли площади колец в падающем пучке:

$$0 \div 0.2\delta g': \quad 0.2^{2/3} - 0^{2/3} = 0.34 = 34\%;$$

$$0.2\delta g' \div 0.4\delta g': \quad 0.4^{2/3} - 0.2^{2/3} = 0.20 = 20\%;$$

$$0.4\delta g' \div 0.6\delta g': \quad 0.6^{2/3} - 0.4^{2/3} = 0.17 = 17\%;$$

$$0.6\delta g' \div 0.8\delta g': \quad 0.8^{2/3} - 0.6^{2/3} = 0.15 = 15\%;$$

$$0.8\delta g' \div 1.0\delta g': \quad 1.0^{2/3} - 0.8^{2/3} = 0.14 = 14\%;$$

If сделить долю площади в падающем кольце на долю площади кольца в фокальной плоскости, то we get quantity пропорциональную интенсивности. Погауается качественно график $I \sim r^{-4/3}$.

Answer: \textbfAnswer: Ring Beam fraction Fraction of circle area 0.0 – 0.2 34% 4% 0.2 – 0.4 20% 12% 0.4 – 0.6 17% 20% 0.6 – 0.8 15% 28% 0.8 – 1.0 14% 36%

C1 0.30 pts

Using the sine theorem for triangles MSC and $MS'C$, as well as Snell's law of refraction, obtain an invariant of the form $n \cdot f(r, s, p) = n' \cdot f(r, s', p') = \text{inv}$.

Theorem of sines for the triangle MSC (see Fig. 2 from the condition):

$$\frac{\sin\theta}{p} = \frac{\sin\angle SMC}{r - s}.$$

Теорема синусов for треangleьника $MS'C$:

$$\frac{\sin\theta}{p'} = \frac{\sin\angle S'MC}{r - s'}.$$

Закон преломления:

$$n\sin\angle SMC = n'\sin\angle S'MC.$$

Let us express from теоремы синусов $\sin\angle SMC$ and $\sin\angle S'MC$, substitute it into the law of refraction, we get invariant

Answer: \textbfAnswer: $\dfrac{n(r-s)}{p} = \text{inv}$

C2 0.10 pts

Using the cosine theorem in triangle MSC, go invariantly from magnitude p to angle θ. Assuming the angle θ to be small, expand the factors in the form of roots in a Taylor series up to the first term with a non-zero degree θ. Divide the resulting approximate value by "$-r$" to obtain the reduced value of the invariant.

Cosine theorem

$$p^2 = r^2 + (r-s)^2 - 2r(r-s)\cos\theta.$$

Using сложение косинуса and полный квадрат

$$p^2 = (r - (r-s))^2 + 2r(r-s)\cdot\frac{\theta^2}{2}.$$

Подставляем в инвариант, we get:

$$\frac{n(r-s)}{\sqrt{s^2 + r(r-s)\theta^2}} = \frac{n(r-s)}{s}\left(1 - \frac{r(r-s)}{2s^2}\right).$$

Разделим на «$-r$", we get the answer.

Answer: \textbfAnswer: $\dfrac{n(r-s)}{\sqrt{s^2 + r(r-s)\theta^2}} = \text{inv}\ n\left(\dfrac{1}{r} - \dfrac{1}{s}\right)\left(1 - \dfrac{r(r-s)\theta^2}{2s^2}\right) = \text{inv}$

C3 — 0.10 pts

Having written the invariant from point C2 for paraxial rays before and after refraction, show the invariance of the quantity $Q(s_0)$.

Let us put $\theta = 0$ in the invariant of clause C2.

Answer: \textbfAnswer: $n\left(\frac{1}{r}-\frac{1}{s_0}\right) = n'\left(\frac{1}{r}-\frac{1}{s'_0}\right) = Q(s) = \text{inv}$

C4 — 0.10 pts

The left-hand side of the invariant obtained in C2 is a function of f of s and θ. Find the value of $f(s,\theta) - f(s_0,\theta)$. Consider δs and θ to be small parameters.

The invariant is a function of s and θ:

$$f(s,\theta) = n\left(\frac{1}{r}-\frac{1}{s}\right)\left(1-\frac{r(r-s)\theta^2}{2s^2}\right).$$

We find value сницы, учитываем что θ одинаковый:

$$f(s,\theta) - f(s_0,\theta) = \frac{\partial f}{\partial s} ds.$$

Since θ тоже малая quantity, то

$$\frac{\partial f}{\partial s} = n\frac{1}{s^2}$$

Answer: \textbfAnswer:

$$\frac{n\delta s}{s_0^2}$$

C5 — 0.20 pts

Using the results of the previous paragraphs, expressing the small angle θ in terms of the radius of curvature of the surface and the small height h, obtain the desired relationship:

$$\frac{n'\delta s'}{s_0'^2} - \frac{n\delta s}{s_0^2} = -\frac{h^2}{2}Q^2(s_0)\left(\frac{1}{n's'_0}-\frac{1}{ns_0}\right).$$

Let's express the angle θ through h:

$$\theta = \frac{h}{r}.$$

Let us represent expression via/through $Q(s)$:

$$\frac{r(r-s)\theta^2}{2s^2} = \frac{(r-s)h^2}{2s^2 r} = n\left(\frac{1}{s}-\frac{1}{r}\right)\frac{h^2}{2ns} = Q(s)\frac{h^2}{2ns}.$$

Следующие величины инвариантны

$$f(s,\theta) = f(s',\theta),$$

$$f(s_0,\theta) + \frac{n\delta s_0}{s_0^2} = f(s'_0,\theta) + \frac{n'\delta s'}{s_0'^2},$$

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$$Q(s_0) - \frac{h^2}{2ns_0}Q^2(s_0) + \frac{n\delta s}{s_0^2} = Q(s'_0) - \frac{h^2}{2n's'_0}Q^2(s'_0) + \frac{n'\delta s'}{s_0'^2}$$

Пользуемся тем, что $Q(s_0) = Q(s_0')$, we get expression

$$\frac{n'\delta s'}{s_0'^2} - \frac{n\delta s}{s_0^2} = -\frac{h^2}{2} Q^2(s_0) \left(\frac{1}{n's_0'} - \frac{1}{ns_0} \right).$$

Answer: \textbfAnswer: CTD

C6 — 0.20 pts

Show that the aberrations of the object and the image after the kth refractive surface are related by the relation:

$$n_k' \, \delta s_k' \, \alpha_k'^2 - n_1 \, \delta s_1 \, \alpha_1^2 = -\frac{1}{2} \sum_{i=1}^{k} h_i^4 \, Q_i^2(s) \left(\frac{1}{n_i' s_i'} - \frac{1}{n_i s_i} \right).$$

Иными словами, for вычисления суммарной аберрации нужно по всем поверхностям сложить правые части доказанного соотношения, умноженного на h_i^2.

Let's move from s_i to the angles on the left side of the formula from C5:

$$\alpha_i = \frac{h_i}{s_i}, \quad \alpha' = \frac{h_i}{s_i'}.$$

For/At прохождении raya между преломляющими поверхностями меняется distance от raya до оптической оси, однако, сохраняется angle наклона:

$$\alpha_i' = \alpha_{i+1},$$

$$n_i' = n_{i+1},$$

$$\delta s_i' = \delta s_{i+1}.$$

Складывая такие выражения

$$n_i' \, \delta s_i' \, \alpha_i'^2 - n_i \, \delta s_i \, \alpha_i^2 = -\frac{h_i^4}{2} Q_i^2(s_i) \left(\frac{1}{n_i' s_i'} - \frac{1}{n_i s_i} \right).$$

on the left side there are only two terms left: one corresponds to the entry of rays into the optical system, the other - entry from it. This makes it possible to calculate aberrations for complex systems.

Answer: \textbfAnswer: CTD

D1 — 2.00 pts

Find the spherical aberration $\delta s_2'$ of a thin converging lens for an incident parallel beam of light. Apply the result of step C6 for two refractive surfaces and the thin lens formula, leaving the dependence only on the radius of curvature of the first surface (where the original beam of light falls). Express your answer in terms of the refractive index n and the reciprocals $\rho = 1/r$ and $\varphi = 1/f'$.

We will use the sign convention as in part C. Then

$$s_1 = \infty, \quad s_2' = -f.$$

Положение изображения после первого преломления:

$$s_1' = s_2 = r\frac{n}{n-1}.$$

Formula тонкой линзы позволяет найти radius кривизны r_2 второй поверхности for/at заданном focuse f and radiuse первой поверхности r:

$$\frac{1}{f} = (n-1)\left(-\frac{1}{r} + \frac{1}{r_2}\right).$$

Since lens тонкая, то $h_1 = h_2 = h$. Левая часть формулы C6:

$$n_2'\,\delta s_2'\,\alpha_2'^2 - n_1\,\delta s_1\,\alpha_1^2 = \delta s_2' \left(\frac{h}{f}\right)^2.$$

$$Q_1 = \frac{1}{r} = \rho.$$

$$Q_2 = \frac{1}{r_2} - \frac{1}{-f} = \frac{n}{n-1}\varphi + \rho.$$

Собираем всё вместе:

$$\delta s_2' = -\frac{h^2}{2\varphi^2}\left(Q_1^2\frac{n-1}{n^2}\rho + Q_2^2\left(-\varphi - \frac{n-1}{n^2}\rho\right)\right)$$

we get the answer.

Answer: \textbfAnswer: $\delta s_2' = \frac{h^2}{2}\left[\frac{\rho^2}{\varphi}\frac{n+2}{n} + \rho\,\frac{2n+1}{n-1} + \varphi\,\frac{n^2}{(n-1)^2}\right]$

D2 0.50 pts

Find the value of ρ^* at which the spherical aberration $\delta s_2'$ is minimal. Express your answer using n and φ.

The expression in square brackets in D1 is a parabola with respect to ρ, the upward branch. The minimum aberration (modulo) will correspond to the vertex. From this we get the value for ρ^*. Note that these arguments are correct only when the parabola does not intersect the horizontal axis. After all, if it crosses, then generally speaking you can get zero aberration. Let's assume it does. Then the discriminant must be non-negative:

$$D = \frac{1-4n}{(n-1)^2} \geq 0,$$

$$n \leq 1/4.$$

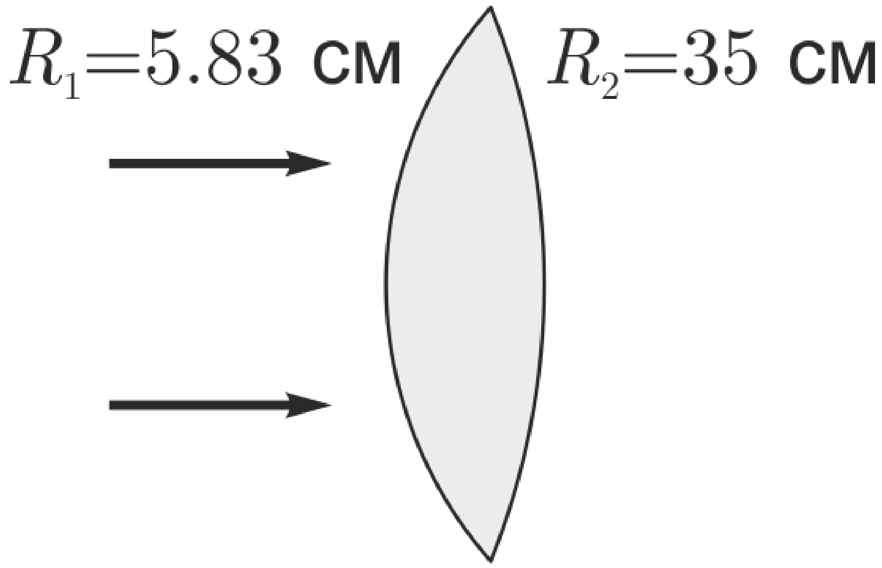
Но for собирающей линзы $n > 1$, therefore, in reality the parabola does not intersect the horizontal axis in our conditions.

Answer: \textbfAnswer: $\rho^* = -\frac{\varphi}{2}\frac{2n+1}{n-1}\frac{n}{n+2}$

D3 0.50 pts

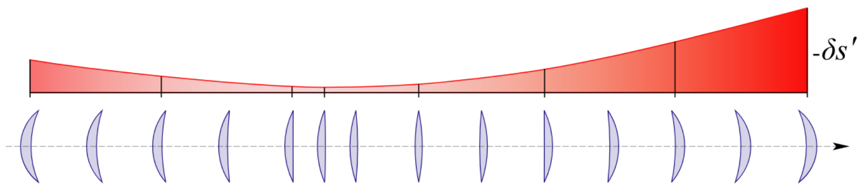
Calculate the radii of curvature of the lens surfaces. Draw (qualitatively) your calculated lens on the answer sheet.

Answer: \textbfAnswer: $R_1 = -5.83\ cm, R_2 = +35\ cm$, i.e. the lens is doubly convex, and the smaller radius of curvature faces the direction of the incident light beam.



D4 0.50 pts

Schematically depict a graph of the modulus of longitudinal spherical aberration $\delta s_2'$ for the lenses shown in Figure 4.



E1 1.50 pts

In what direction and at what distance should the observation plane be shifted from the focal plane so that the scattering circle is of minimum size?

The fact from part B is used that the aberration is quadratic in the distance of the beam from the optical axis h:

$$\delta s = \frac{h^2}{\varepsilon}, \quad \delta s' = \frac{H^2}{\varepsilon},$$

where ε – некий coefficient (смерности длины). Here H – radius линзы (максимально удалённый ray), h – произвольный ray. Similarly, $\delta s'$ – максимальная аберрация, δs – точка пересечения оптической оси rayем h. If что тут δs and $\delta s'$ положительные, отчитываются от focusa в сторону линзы. Distance r от оси до точки rayа на расстоянии x от focusa, который до преломления идёт на расстоянии h от оси можно найти from подобия (рис. E1):

$$\frac{r}{x - \delta s} = \frac{h}{f}.$$

Погауается что r есть function плоскости наблюдения x and rayа h:

$$r(x, h) = \frac{1}{f}\left(xh - \frac{h^3}{\varepsilon}\right).$$

We find, for/at каком h for/at некотором фиксированном x будет максимум

$$h_1 = \sqrt{\frac{\varepsilon}{3}} x^{1/2}.$$

Максимальный radius пучка в зависимости от плоскости наблюдения

$$r_{\max}(x) = \frac{2\varepsilon^{1/2}x^{3/2}}{3^{3/2}f}.$$

Это погауается некоторая огибающая, которую мы we get, смотря на rayand, которые не пересекают оптическую ось. Чем ближе к focusy, тем меньше смер кружка, который дают эти rayand. Однако, ещё есть rayand, которые пересекают оптическую ось and for/at for/atближении к focusy увеличивают смер кружка. Зависимость расстояния от оси до точки raya for крайнего противоположного raya от координаты

$$y(x) = \frac{H}{f}(\delta s' - x).$$

Минимальный кружок будет there, where эти эффекты equal $r_{\max}(x) = y(x)$:

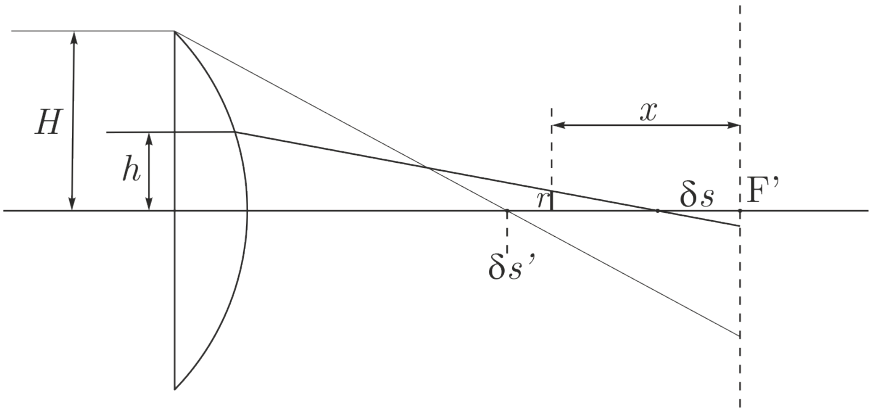
$$\frac{2\varepsilon^{1/2}x^{3/2}}{3^{3/2}f} = \frac{H}{f}(\delta s' - x),$$

$$\frac{2}{3^{3/2}}\left(\frac{x}{\delta s'}\right)^{3/2} = \left(1 - \frac{x}{\delta s'}\right).$$

Решая это equation, we get answer

$$x = \frac{3}{4}\delta s'.$$

Answer: \textbfAnswer: To the lens at a distance $\frac{3}{4}|\delta s'|$.



E2 **1.50 pts**

Draw a schematic graph of light intensity versus radial coordinate in the minimum scattering circle. To do this, follow steps similar to point B4.

Let us first answer the question: how does the radius of the circle (r) in the E1 plane relate to the radius of the circle (h) in the beam incident on the lens? From the previous paragraph it turns out that they are connected as follows:

$$r = h|3 - 4h^2|,$$

here r – отнормировано на maximum radius пятна $\left(\frac{1}{4}\frac{\delta s' H}{f}\right)$ в рассматриваемой плоскости, h – на radius пучка H. В общем crayae одному radiusy в плоскости наблюдения соответсвует целых три radiusa в падающем пучке. Доля площади кружка оказывается такой же, как and в прошлый times. Чтобы найти долю пучка, нужно сложить долю площади трех колец, that is величины вида $h_2^2 - h_1^2$. For нахождения h_1 and h_2 of each ring you need to solve an equation of the third degree.

Answer: \textbfAnswer: Ring Beam fraction Fraction of circle area Intensity 0.0 – 0.2 12% 4% 3.01 0.2 – 0.4 13% 12% 1.11 0.4 – 0.6 15% 20% 0.76 0.6 – 0.8 19% 28% 0.67 0.8 – 1.0 41% 36% 1.13

плоскость наблюдения			первое кольцо		второе кольцо		третье кольцо			
r2	r1	r2^2-r1^2	h2	h1	h2	h1	h2	h1	доля пучка	интенсивность
1	0,8	0,36	0,5	0,3042	0,6729	0,5	1	0,9771	0,40553236	1,12647878
0,8	0,6	0,28	0,3042	0,2129	0,7397	0,6729	0,9771	0,9526	0,18885056	0,67446629
0,6	0,4	0,2	0,2129	0,1367	0,7895	0,7397	0,9526	0,9262	0,152394	0,76197
0,4	0,2	0,12	0,1367	0,0671	0,8305	0,7895	0,9262	0,8976	0,13276516	1,10637633
0,2	0	0,04	0,0671	0	0,8660	0,8305	0,8976	0,8660	0,12045792	3,011448